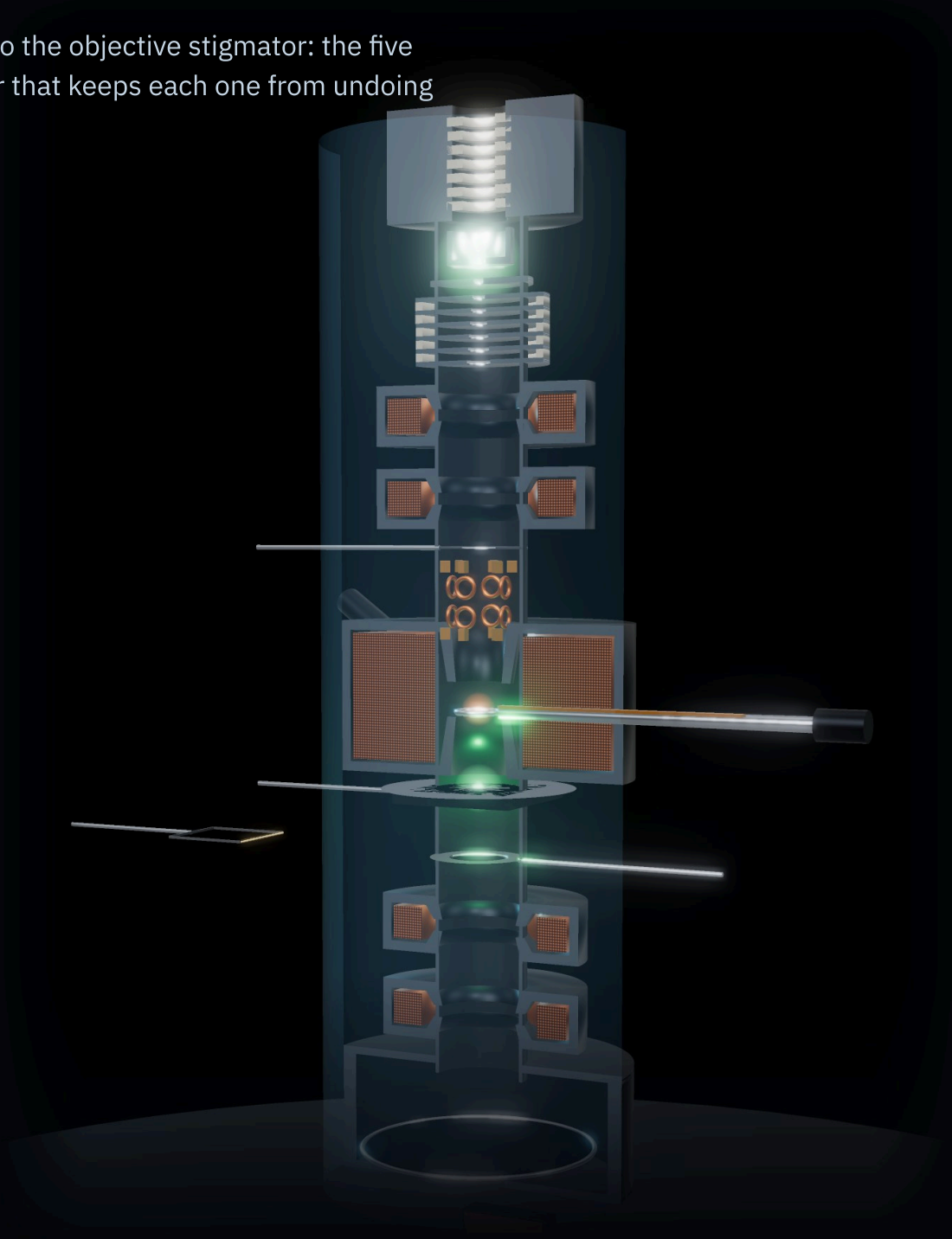
TEM direct *alignment*

From the electron gun to the objective stigmator: the five alignments, in the order that keeps each one from undoing the last.



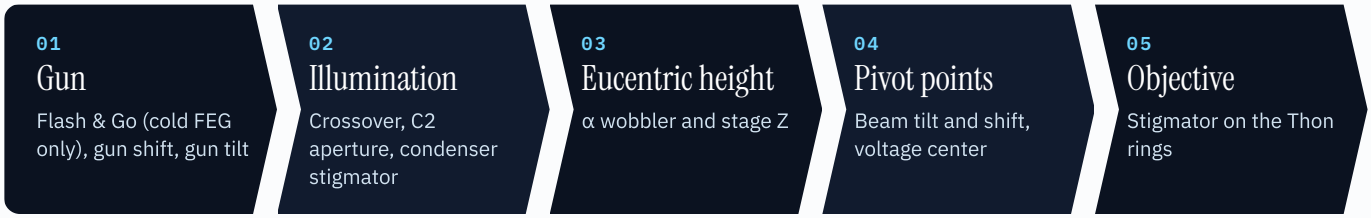
SOP-01 · version 1.0 · October 2026
Level: new users and refreshers
Practice it on the TEM-view digital twin

Seeing with Electrons
TEM-VIEW · ADVANCED-EM

BEFORE YOU START

Why the *order* matters

Each alignment assumes the ones before it are done. Correct the condenser stigmator with an off-center C2 aperture, or align pivot points with the specimen at the wrong height, and later corrections drift away. Work from the gun down the column, and go back a step whenever a cue looks wrong.



Check first

- [ ] Column and gun vacuum are in the green; column valves are open only when you are ready to view.
- [ ] High tension is at the operating voltage and has had time to settle.
- [ ] The specimen is inserted, the holder is seated, and the stage is at $\alpha = \beta = 0°$.
- [ ] The objective aperture is out; the selected-area aperture is out.
- [ ] You are in TEM mode at a medium magnification with the beam on the fluorescent screen (or a screen-view camera).
- [ ] You know which electron source your microscope has (see right).

VOCABULARY

Brightness (also called intensity or C2) spreads or focuses the beam. **Crossover** is the setting where the beam is smallest. **Multifunction knobs** are the X/Y knobs that drive whichever alignment is selected.

Know your electron source

SOURCE	EMISSION	FLASH & GO
Cold FEG room-temperature W tip	Falls slowly as gas adsorbs on the tip	**Yes**: when emission has dropped, and at the start of a session
Schottky FEG ZrO/W tip at ~1800 K	Steady for months	**No**: never flash a Schottky source
LaB_6 / W thermionic	Set by filament heating and saturation	**No**: saturate the filament instead

FLASH & GO IS FOR COLD FEG ONLY

The flash heats a cold-field-emission tip for a moment to drive off adsorbed gas. A Schottky or thermionic gun runs hot all the time and has no flash. If your gun is not a cold FEG, skip that part of step 1.

NAMES VARY BY VENDOR

This guide uses generic names. Menus, knob labels and wobbler names differ between manufacturers and software versions. Your facility's procedures and the manufacturer's manual come first.

1 GUN

Gun: emission, *shift* and *tilt*

The gun alignments put the source on the axis of the condenser system. When they are off, the beam walks across the screen every time you change spot size, and you lose current.

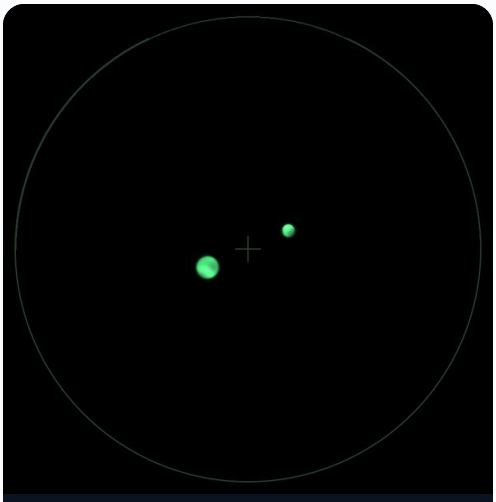
Procedure

1. **Cold FEG only:** if the emission current has dropped (or at the start of the day), run **Flash & Go**. Wait for the emission to settle before aligning. Skip this on a Schottky or thermionic gun.
2. Bring the beam to **crossover** with the brightness knob, so it is a small spot.
3. Select a **large spot number** (the smallest spot). Center the spot on the screen with **beam shift**.
4. Select a **small spot number** (a larger spot). If the spot has moved, center it with **gun shift**, not beam shift.
5. Repeat steps 3 and 4 until the spot stays put as you change spot size.
6. Select **gun tilt** and adjust it to **maximize the screen current** (or the brightness of the spot).

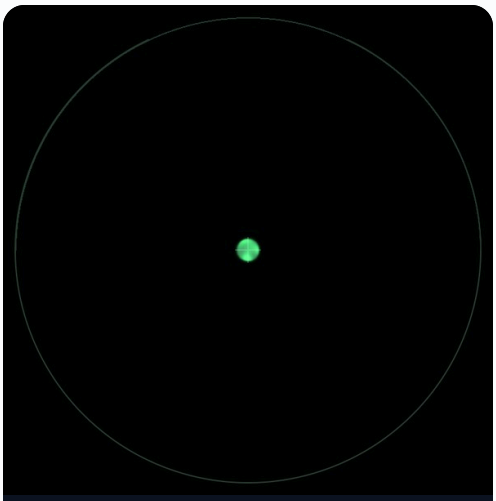
✓ The crossover stays centered at every spot size, and the screen current is at its maximum.

WHY TWO KNOBS

Beam shift moves the beam below the condenser lenses; gun shift moves it at the source, and how far a source offset moves the spot depends on the spot size. Correcting at each end of the spot range and iterating converges on both. If your direct-alignment software prompts for a different spot size or knob, follow its prompts.



OFF Two spot sizes superimposed: the crossover **walks** when the spot size changes.



ALIGNED Both spot sizes land on the **same place**.

2 ILLUMINATION

Crossover, *C2 aperture*, condenser stigmator

A centered C2 aperture makes the beam open and close about one point, so changing brightness never moves the illuminated area. A corrected condenser stigmator keeps the beam round on both sides of focus.

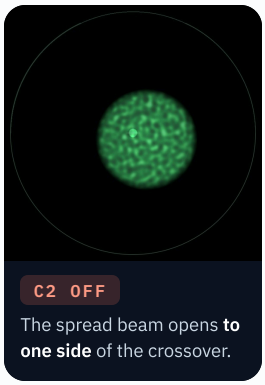
C2 aperture

1. At crossover, center the spot with **beam shift**.
2. Spread the beam with the brightness knob until it fills about half the screen.
3. If it does not expand **concentrically** about the center, move the **C2 aperture** (mechanical or motorized) toward the center.
4. Return to crossover, re-center with beam shift, spread again. Repeat until the beam opens about one point.

Condenser stigmator

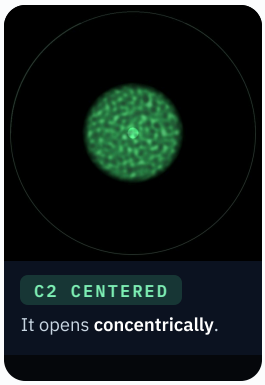
1. Go a little to **one side of crossover**: an astigmatic beam is an ellipse.
2. Go to the **other side**: the ellipse turns by 90°. That flip is condenser astigmatism.
3. Select the **condenser stigmator** and adjust X and Y until the beam is **round** on both sides of crossover.

✓ The beam expands concentrically about the screen center, and stays round on both sides of focus.



C2 OFF

The spread beam opens **to one side** of the crossover.



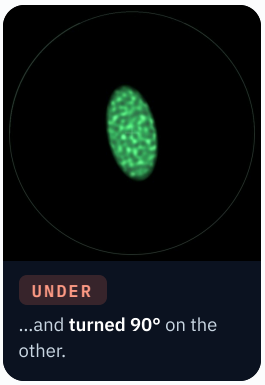
C2 CENTERED

It opens **concentrically**.



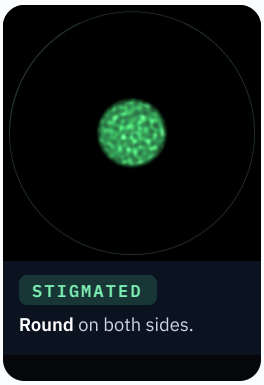
OVER

Elliptical on one side of crossover...



UNDER

...and **turned 90°** on the other.



STIGMATED

Round on both sides.

COMMON MISTAKES

Moving the C2 aperture while the beam is spread, without first re-centering the crossover with beam shift: the two corrections then chase each other. Do one at crossover and the other spread, and alternate.

CHECK THE FLIP

Judging condenser astigmatism on one side of focus only can hide it: a slightly elliptical beam may just be off-round from the aperture edge. The 90° flip through crossover is the reliable sign.

3 EUCENTRIC HEIGHT

Put the specimen on the *tilt axis*

At the eucentric height, the region you are looking at stays in place when the stage tilts. It is also the height the objective lens is designed for, so focus, magnification and the later alignments are all defined there.

Procedure

1. Find a recognizable feature (an edge, a particle) at a medium magnification and center it.
2. Turn on the **α wobbler** (Z-wobbler). The stage rocks back and forth, typically by ±5° to ±15°.
3. If the image **swings sideways**, adjust **stage Z** until the movement stops. Large swings: big Z steps; small swings: fine steps.
4. At the right height the feature **only pulses in place**. Turn the wobbler off.
5. Optionally, set the objective to its standard focus (eucentric focus) and refocus with Z, if that is your facility's practice.

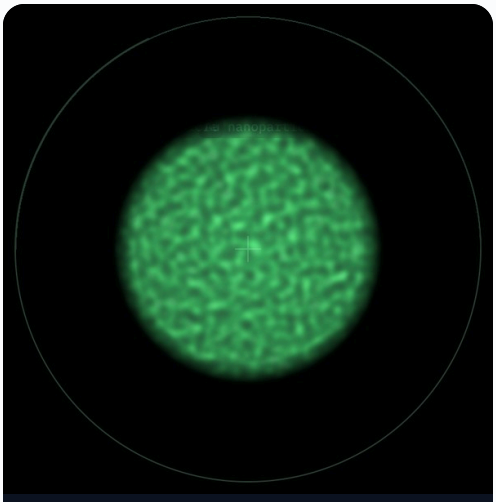
✓ With the α wobbler on, the image no longer shifts laterally: it pulses in place.

TIP

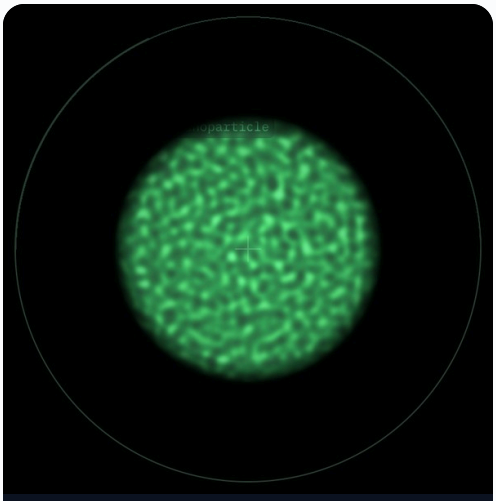
Re-check the eucentric height whenever you move to a new area of a specimen that is not flat, and after re-inserting the holder.

IF IT NEVER STOPS MOVING

Make sure the wobbler is tilting α (not β) and that the image shift you see is not drift: turn the wobbler off and watch for a few seconds. If the stage Z reaches its limit, the holder or specimen may be mounted too high or too low.



WRONG Z Two moments of the wobble averaged: the image **smears** as it swings sideways.



EUCENTRIC The same two moments overlap: the image **stays sharp**.

4 PIVOT POINTS & VOLTAGE CENTER

Tilt without *moving*

Pivot points make beam tilt and beam shift independent: tilting the beam must not translate it across your region of interest, and shifting it must not tilt it. The voltage (or current) center then puts the beam on the objective lens's axis.

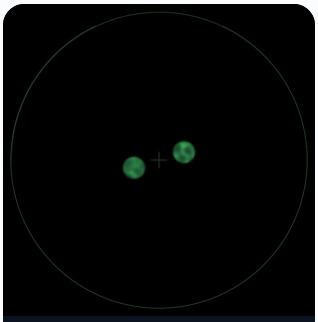
Pivot points

1. Near crossover with a small spot, select **beam tilt pivot point X**. The wobbler tilts the beam back and forth in X.
2. If you see **two spots** (or one swinging spot), merge them into one with the multifunction knobs.
3. Repeat for **pivot point Y**.
4. If your instrument lists them, do the same for the **beam shift pivot points** X and Y.

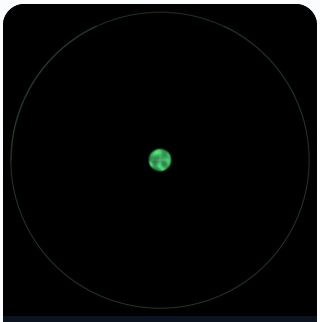
Voltage center (rotation center)

1. Spread the beam and find a feature near the screen center.
2. Turn on the **HT wobbler** (voltage center), or the objective-current wobbler (current center), as your instrument offers.
3. The image pulses. If it **swings** to one side, adjust **beam tilt** with the multifunction knobs until it pulses **symmetrically about the center**.

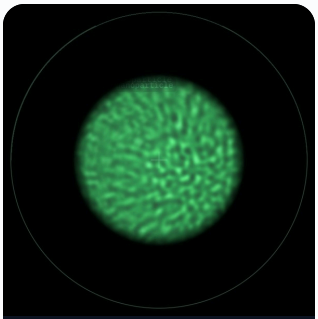
✓ The pivot-point wobblers show one stationary spot, and the HT wobbler makes the image breathe in place.



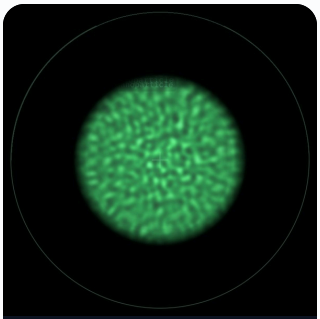
OFF Two spots: tilting **translates** the beam.



ALIGNED One spot: a **pure tilt**.



OFF-AXIS The HT wobble **swings** the image: averaged, it smears.



CENTERED It pulses about the center.

FOR HIGH RESOLUTION

The voltage center is a good starting point. For atomic-resolution work, follow it with a coma-free alignment (beam-tilt-induced image shift or a Zemlin tableau), with your facility's procedure.

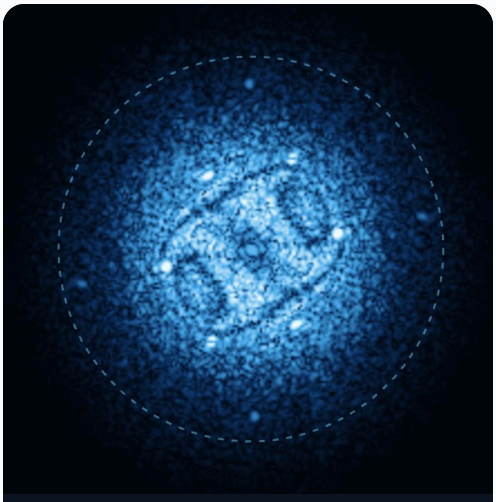
5 OBJECTIVE STIGMATOR

Round *Thon rings*

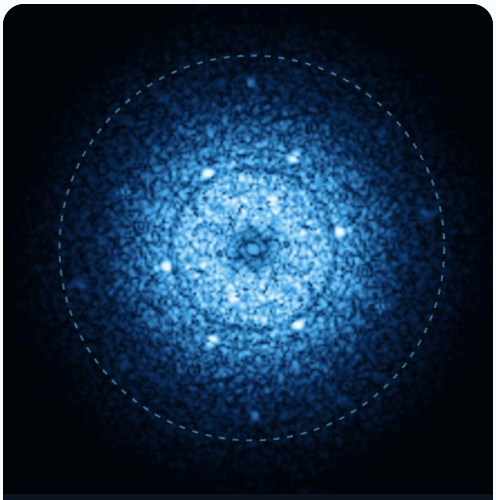
Objective astigmatism focuses one direction more strongly than the other. In the live FFT (diffractogram) of an amorphous area, it turns the circular Thon rings into ellipses, or into hyperbolas close to focus.

1. Move to a thin **amorphous** area (carbon film or the edge of the specimen) at high magnification.
2. Turn on the **live FFT**. Go somewhat **underfocus** so several rings are visible.
3. Select the **objective stigmator**. Adjust X, then Y, until the rings are **circular**.
4. Check through focus: the rings should shrink and grow evenly, without stretching in one direction.
5. Return to your working focus and area.

✓ The Thon rings are circular at every defocus.



ASTIGMATIC Rings stretched into a **bow tie**.



CORRECTED **Circular** rings.

Quick reference

	ALIGNMENT	DONE WHEN	INITIALS
☐	Flash & Go (cold FEG only)	Emission recovered and stable	
☐	Gun shift / gun tilt	Spot fixed across spot sizes; current maximal	
☐	C2 aperture	Beam opens concentrically	
☐	Condenser stigmator	Round on both sides of crossover	
☐	Eucentric height	No lateral shift with the α wobbler	
☐	Pivot points X, Y	One stationary spot	
☐	Voltage center	Image pulses symmetrically	
☐	Objective stigmator	Circular Thon rings	

PRACTICE BEFORE YOUR SESSION

Try it on the *digital twin*

Every figure in this guide was computed by TEM-view, a physics-based transmission electron microscope that runs in your browser. Its training task **“TEM: direct alignments”** hides a different misalignment at each of these steps every time. At the end, it shows you how close you got.

OPEN THE TRAINING TASK

advanced-em.github.io/TEM-view/?train=tem-direct-align

On a phone, scan the code. On a computer, open the link, or open TEM-view and press **Training**.

SEE THE PHYSICS

Click any part of the 3D column to learn what it does and how it is measured.

GET YOUR TRUE SCORE

Rate your alignment, then see the hidden answer and each check, pass or fail.

HELP BUILD BETTER MICROSCOPES

With your consent, anonymous practice sessions train AI that can assist real alignments.